

LIVE DEMO SYSTEM

Interactive Interface Components

Component	Functionality
Input Selection	A dropdown menu allowing the user to select between the pre-uploaded stems (Vocals, Bass, Drums, Other) or an "Upload" button for custom .wav files.
Real-time Processing	A Bitrate Slider (ranging from 32 kbps to 320 kbps). When the slider moves, the system triggers the compression script in the background to generate a new compressed buffer.
Comparative Playback	Two synchronized audio players: Original (Lossless) vs. Compressed . A "Toggle" button allows A/B testing during playback to hear artifacts.
Live Visualizer	A dynamic Spectrogram Comparison that updates based on the selected bitrate, highlighting the "Frequency Cutoff" typical of lossy compression.

Demo Workflow Example

1. **User Input:** The user uploads vocals.wav.
2. **Parameter Adjustment:** The user slides the bitrate down to **64 kbps**.
3. **Processing:** The system calculates the **SNR** (e.g., 18.5 dB) and **Compression Ratio** (e.g., 22:1) instantly.
4. **Output Visualization:** * The **Spectrogram** shows a clear horizontal line where high-frequency data is truncated.
 - The user clicks **Playback** to hear the "metallic" artifacts in the vocal sibilance.